Probability Theory and Random Processes (MA225)

Lecture Slides

Lecture 24

Hitting Time

Def: For any $A \subset S$ (state space), the hitting time T^A is defined by

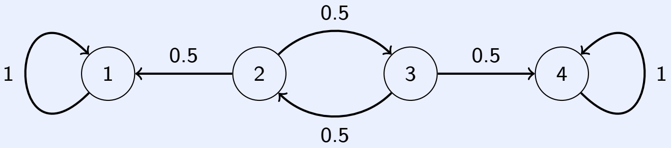
$$T^A = \inf\{n \geq 0 : X_n \in A\},$$

with the convention that $\inf \emptyset = \infty$.

- If $X_0 \in A$, then $T^A = 0$.
- If $X_0 \notin A$, then T^A is the first time after 0, when the chain enters A.
- $T^{\{i\}}$ will be denoted by T^i, $i \in S$.
- The probability that starting from i, the MC will ever hit A is given by $h_i^A = P_i\left(T^A < \infty\right)$.
- Informally, we will write $h_i^A = P_i(\text{hit } A)$.
- If A is a closed class, then h_i^A is called **absorption probability**.

Example

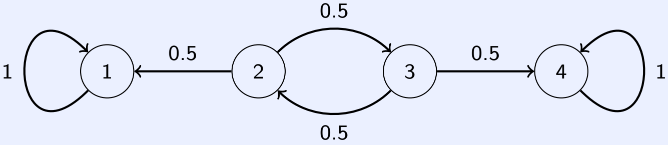
Example 1: Consider the MC with one-step transition probability is specified by the following diagram.



Starting from 2, what is the probability of absorption in 4?

- Denote $h_i = P_i(\text{hit } 4)$.
- Easy to see that $h_1 = 0$ and $h_4 = 1$.

Example



- Conditioning on the first transition, we get

$$h_2 = \frac{1}{2}h_1 + \frac{1}{2}h_3,$$
$$h_3 = \frac{1}{2}h_2 + \frac{1}{2}h_4.$$

- Thus,

$$h_2 = \frac{1}{2}h_3 = \frac{1}{2}\left(\frac{1}{2}h_2 + \frac{1}{2}\right) \implies h_2 = \frac{1}{3}.$$

Hitting Probability

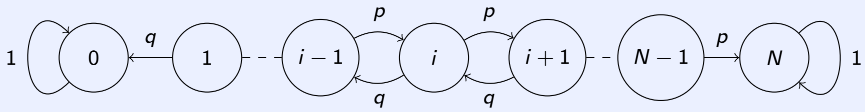
Theorem: The vector of hitting probabilities $h^A = \left(h_i^A : i \in S\right)$ is the minimal non-negative solution of the system of linear equations

$$\begin{aligned} h_i^A &= 1 \quad \text{for } i \in A \\ h_i^A &= \sum_{j \in S} p_{ij} h_j^A \quad \text{for } i \notin A. \end{aligned}$$

Here minimality means that if $x = (x_i : i \in S)$ is another solution with $x_i \geq 0$ for all i, then $x_i \geq h_i^A$ for all i.

Example

Example 2: [Gambling Model] Consider the Gambler ruin problem, where the MC is given by the following diagram with $q = 1 - p \in (0, 1)$.

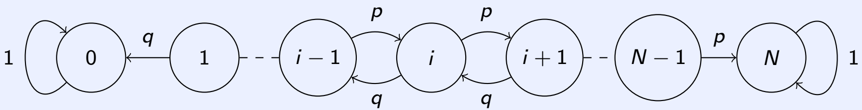


Thus, the one-step transition probabilities are

$$
\begin{aligned}
&p_{00} = p_{NN} = 1 \\
&p_{i,i+1} = p,\ p_{i,i-1} = q \quad \text{for} \quad i = 1,\ 2,\ \ldots,\ N-1 \\
&p_{ij} = 0 \quad \text{otherwise.}
\end{aligned}
$$

What is the probability that the gambler starting with an initial fortune of i goes back bankrupt?

Example



Let $h_i = P_i\,(\text{hit } 0)$. Then h is the minimal non-negative solution to

$$\begin{aligned} h_0 &= 1 \\ h_i &= ph_{i+1} + qh_{i-1} \quad \text{for } i = 1, 2, \ldots, N-1. \end{aligned}$$

Since, $0 \leq h_N \leq 1$ and we are interested in minimal non-negative solution, we must take $h_N = 0$.

Example

Now, we have

$$h_{N-2} - h_{N-1} = \frac{p}{q}\left(h_{N-1} - h_N\right) = \frac{p}{q} h_{N-1}$$

$$h_{N-3} - h_{N-2} = \frac{p}{q}\left(h_{N-2} - h_{N-1}\right) = \left(\frac{p}{q}\right)^2 h_{N-1}$$

$$\vdots$$

$$h_i - h_{i+1} = \frac{p}{q}\left(h_{i+1} - h_{i+2}\right) = \left(\frac{p}{q}\right)^{N-i-1} h_{N-1}$$

$$\vdots$$

$$h_0 - h_1 = \frac{p}{q}\left(h_1 - h_2\right) = \left(\frac{p}{q}\right)^{N-1} h_{N-1}.$$

Example

For $i = 0, 1, 2, \dots, N-2$, adding first $N-i-1$ equations, we obtain

$$h_i - h_{N-1} = \left[\frac{p}{q} + \dots + \left(\frac{p}{q}\right)^{N-i-1}\right] h_{N-1}$$

$$\implies h_i = \left[1 + \frac{p}{q} + \dots + \left(\frac{p}{q}\right)^{N-i-1}\right] h_{N-1}$$

$$\implies h_i = \begin{cases} \frac{1-(p/q)^{N-i}}{1-(p/q)} h_{N-1} & \text{if } p \neq \frac{1}{2} \\ (N-i)\, h_{N-1} & \text{if } p = \frac{1}{2}. \end{cases}$$

Since, $h_0 = 1$, we get

$$h_i = \begin{cases} \frac{1-(p/q)^{N-i}}{1-(p/q)^N} & \text{if } p \neq \frac{1}{2} \\ 1 - \frac{i}{N} & \text{if } p = \frac{1}{2}. \end{cases}$$

Expected Hitting Time

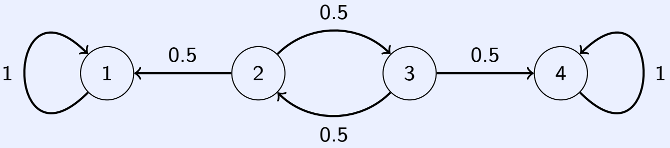
- Starting from i, the expected (mean) time taken by the MC to hit A is given by

$$k_i^A = E_i\left(T^A\right) = \sum_{n \geq 1} nP_i(T^A = n) + \infty P_i\left(T^A = \infty\right).$$

- k_i^A is also called expected (mean) hitting time to hit A starting from i.
- Informally, we will write $k_i^A = E_i(\text{time to hit } A)$.

Example

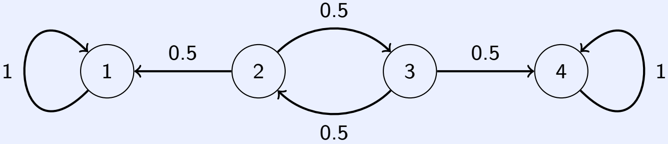
Example 3: Consider the MC with one-step transition probability is specified by the following diagram.



Starting from 2, what is the mean time taken by the chain to get absorbed in 1 or 4?

- Denote $k_i = E_i\,(\text{time to hit } \{1, 4\})$.
- Easy to see that $k_1 = k_4 = 0$.

Example



- Again, conditioning on the first transition, we get

$$k_2 = \frac{1}{2}(1 + k_1) + \frac{1}{2}(1 + k_3) = 1 + \frac{1}{2}k_1 + \frac{1}{2}k_3,$$
$$k_3 = \frac{1}{2}(1 + k_2) + \frac{1}{2}(1 + k_4) = 1 + \frac{1}{2}k_2 + \frac{1}{2}k_4.$$

- Thus, by solving, we get

$$k_2 = 1 + \frac{1}{2}k_3 = 1 + \frac{1}{2}\left(1 + \frac{1}{2}k_2\right) \implies k_2 = 2.$$

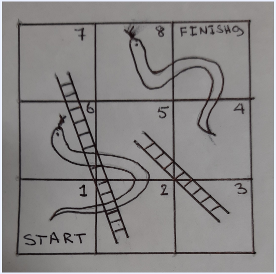
Mean Hitting Times

Theorem: The vector of mean hitting times $k^A = \left(k_i^A : i \in S\right)$ is the minimal non-negative solution of the system of linear equations

$$\begin{aligned} k_i^A &= 0 \quad \text{for } i \in A \\ k_i^A &= 1 + \sum_{j \notin A} p_{ij} k_j^A \quad \text{for } i \notin A. \end{aligned}$$

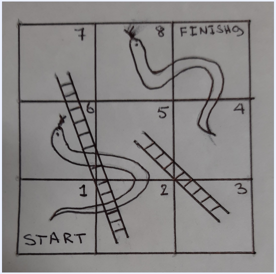
Example

Example 4: Consider a simple game of snakes-and-ladders played on a 3×3 board.



At each turn a player tosses a fair coin and advances one or two places according to whether the coin lands heads or tails. If you land at the foot of a ladder you climb to the top, but if you land at the head of a snake you slide down to the tail. How many turns on average does it take to complete the game?

Example



The TPM is given by

$$P = \begin{array}{c} \\ 1 \\ 4 \\ 5 \\ 7 \\ 9 \end{array} \begin{array}{c} \begin{array}{ccccc} 1 & 4 & 5 & 7 & 9 \end{array} \\ \begin{bmatrix} 0 & 0 & 0.5 & 0.5 & 0 \\ 0.5 & 0 & 0.5 & 0 & 0 \\ 0.5 & 0 & 0 & 0.5 & 0 \\ 0 & 0.5 & 0 & 0 & 0.5 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

Example

Let $k_i = E_i\,$(time to hit 9). Thus we want to find the minimal non-negative solution of

$$
\begin{aligned}
k_1 &= 1 + \frac{1}{2}k_5 + \frac{1}{2}k_7 \\
k_4 &= 1 + \frac{1}{2}k_1 + \frac{1}{2}k_5 \\
k_5 &= 1 + \frac{1}{2}k_1 + \frac{1}{2}k_7 \\
k_7 &= 1 + \frac{1}{2}k_4 + \frac{1}{2}k_9 \\
k_9 &= 0.
\end{aligned}
$$

$$
P = \begin{array}{c} \\ 1 \\ 4 \\ 5 \\ 7 \\ 9 \end{array}
\begin{array}{c}
\begin{array}{ccccc} 1 & 4 & 5 & 7 & 9 \end{array} \\
\begin{bmatrix}
0 & 0 & 0.5 & 0.5 & 0 \\
0.5 & 0 & 0.5 & 0 & 0 \\
0.5 & 0 & 0 & 0.5 & 0 \\
0 & 0.5 & 0 & 0 & 0.5 \\
0 & 0 & 0 & 0 & 1
\end{bmatrix}
\end{array}
$$

Solving, we get $k_1 = 6.625$.